

## Topic 9 — Time Series: White Noise, AR, MA, ARMA and ARIMA Models

## What Is a Time Series?

### Definition

A **time series** is an infinite sequence of random numbers parametrized (indexed) by (discrete) time:  $X_1, X_2, X_3, \dots, X_n, \dots \in \mathbb{R}$ . In real data, the values  $X_1, X_2, \dots$  are typically **not independent**.

Example:  $X_n = 1.005 X_{n-1} + Y_n$ , where  $Y_n$  is random and may itself depend on time  $n$ .

# Stationarity

## Definition

A time series  $\{X_n\}$  is called **(weakly) stationary** if:

1.  $\mathbb{E}(X_n)$  does not depend on  $n$  (constant);
2.  $\mathbb{E}(X_n^2) < \infty$  for all  $n$ ;
3.  $\text{cov}(X_n, X_{n+m}) = \mathbb{E}(X_n X_{n+m}) - \mathbb{E}(X_n) \cdot \mathbb{E}(X_{n+m})$  does **not** depend on  $n$  — it depends on  $m$  only.

## Remark

In other words,  $\text{cov}(X_n, X_\ell)$  depends only on the time-lag  $\ell - n$ . If  $\{X_n\}$  is stationary, then  $\text{Var}(X_n) = \text{cov}(X_n, X_n)$  does not depend on  $n$  either, and neither does  $\text{corr}(X_n, X_\ell)$  (as a function of  $\ell - n$ ).

## Example: A Stationary Time Series

### Example

Let  $Y, Z$  be uncorrelated,  $\mathbb{E}(Y) = \mathbb{E}(Z) = 0$ ,  $\text{Var}(Y) = \text{Var}(Z) = \sigma^2$  (e.g. independent  $N(0, \sigma^2)$ ). For fixed  $\lambda \in [0, 2\pi]$ , define

$$X_n = Y \cos(n\lambda) + Z \sin(n\lambda).$$

Then  $\mathbb{E}(X_n) = \cos(n\lambda)\mathbb{E}(Y) + \sin(n\lambda)\mathbb{E}(Z) = 0$  — constant in  $n$ .

### Example

Using  $\mathbb{E}(Y^2) = \mathbb{E}(Z^2) = \sigma^2$  and  $\mathbb{E}(YZ) = \text{cov}(Y, Z) = 0$ , a direct expansion gives

$$\mathbb{E}(X_n X_{n+m}) = (\cos(n\lambda) \cos((n+m)\lambda) + \sin(n\lambda) \sin((n+m)\lambda)) \sigma^2 = \cos(m\lambda) \sigma^2.$$

So  $\mathbb{E}(X_n^2) = \sigma^2 < \infty$  and  $\text{cov}(X_n, X_{n+m}) = \cos(m\lambda) \sigma^2$  depends on  $m$  only:  $\{X_n\}$  is **stationary**.

## White Noise

### Definition

A time series  $\{Z_n\}$  is a **white noise** if (1)  $\mathbb{E}(Z_n) = 0$ , and (2) for some  $\sigma > 0$ ,

$$\text{cov}(Z_n, Z_{n+m}) = \begin{cases} \sigma^2, & m = 0, \\ 0, & m \neq 0. \end{cases}$$

I.e. white noise has zero mean and uncorrelated values.

### Remark

The standard example of white noise: i.i.d.  $Z_n \sim \mathcal{N}(0, \sigma^2)$ .

## Autoregressive Model AR(1)

### Example

$$X_n = \mu + \alpha(X_{n-1} - \mu) + Z_n, \quad X_0 = a \in \mathbb{R},$$

where  $\mu, \alpha$  are constants and  $\{Z_n\}$  is white noise, independent of  $\{X_n\}$ . Then  $X_1 = \mu + \alpha(a - \mu) + Z_1$ ,  $X_2 = \mu + \alpha(X_1 - \mu) + Z_2$ , and so on — each new value uses a fresh, independent  $Z_n$ .

Writing  $Y_n := X_n - \mu$  gives  $Y_n = \alpha Y_{n-1} + Z_n$ . We look for conditions making  $\{X_n\}$  (hence  $\{Y_n\}$ ) stationary.

## AR(1): Finding the Stationarity Condition

If  $\mathbb{E}(Y_n) = k$  (constant), then  $k = \alpha k + 0$ , so  $\alpha = 1$  or  $k = 0$ . Since  $\alpha = \pm 1$  would force  $\sigma = 0$  (impossible), we need  $\alpha \neq \pm 1$  and  $k = 0$ , i.e.  $\mathbb{E}(X_n) = \mu$ .

If  $\text{Var}(Y_n) = v$  (constant), independence of  $Z_n$  gives  $v = \alpha^2 v + \sigma^2$ , i.e.  $v(1 - \alpha^2) = \sigma^2$ , so

$$\text{Var}(X_n) = v = \frac{\sigma^2}{1 - \alpha^2}.$$

Since  $\text{Var}(X_n) \geq 0$ , we need  $1 - \alpha^2 > 0$ , i.e.  $|\alpha| < 1$ .

### Definition

$|\alpha| < 1$  is necessary and sufficient for stationarity of  $X_n = \alpha X_{n-1} + Z_n$ . Equivalently, using the **characteristic equation**  $1 - \alpha\lambda = 0$  (so  $\lambda = \frac{1}{\alpha}$ ): the series is stationary iff  $|\lambda| > 1$ .

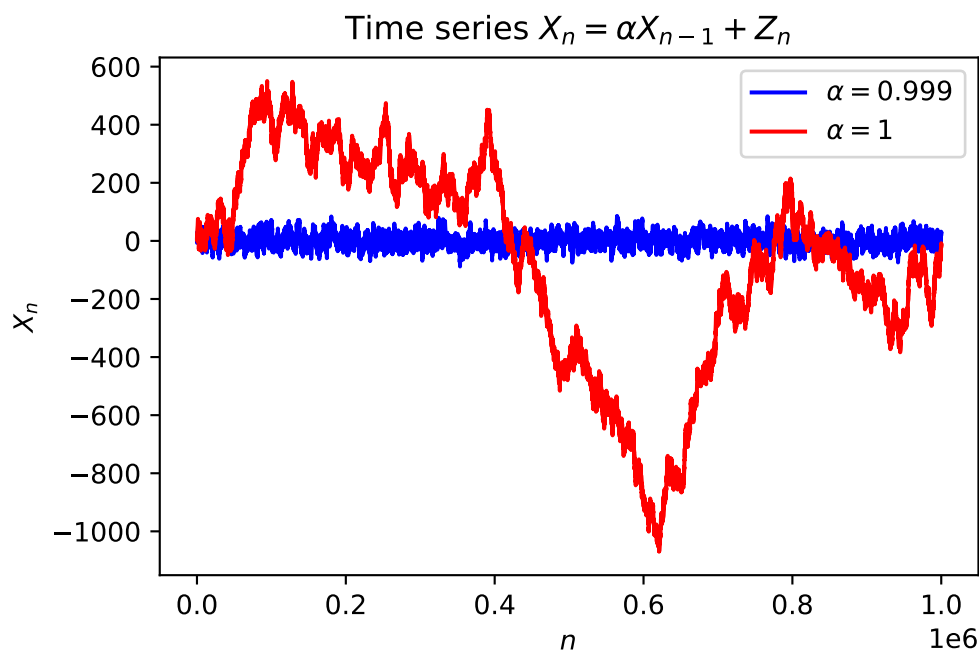


Figure 1: Qualitative difference between stationary (blue) and non-stationary (red) time series behaviour

## Autoregressive Model AR(2)

### Example

$$X_n = \mu + \alpha_1(X_{n-1} - \mu) + \alpha_2(X_{n-2} - \mu) + Z_n, \quad \alpha_2 \neq 0,$$

with initial values  $X_0 = a$ ,  $X_1 = b$ . Then  $X_2 = \mu + \alpha_1(b - \mu) + \alpha_2(a - \mu) + Z_2$ , then  $X_3$  uses  $X_2$  and  $X_1$ , and so on. With  $Y_n = X_n - \mu$ :  $Y_n = \alpha_1 Y_{n-1} + \alpha_2 Y_{n-2} + Z_n$ .

### Definition

The **characteristic equation** is  $1 - \alpha_1 \lambda - \alpha_2 \lambda^2 = 0$ , with two roots  $\lambda_1, \lambda_2$  (possibly complex). The time series is **stationary** iff  $|\lambda_1| > 1$  **and**  $|\lambda_2| > 1$ .



## Reminder: Quadratic Equations and Complex Numbers

### Remark

$a\lambda^2 + b\lambda + c = 0$  ( $a \neq 0$ ) has roots  $\lambda_{1,2} = \frac{-b \pm \sqrt{D}}{2a}$ , discriminant  $D = b^2 - 4ac$ . If  $D < 0$ , the roots are complex:  $\lambda_{1,2} = p \pm qi$  ( $i^2 = -1$ ), with  $|p \pm qi| = \sqrt{p^2 + q^2}$ .

## Example 1: AR(2), Stationary

### Example

$$X_n = \frac{1}{12}X_{n-1} + \frac{1}{2}X_{n-2} + Z_n.$$

Characteristic equation:  $\lambda^2 + \frac{1}{6}\lambda - 2 = 0$ ,  $D = \frac{1}{36} + 8 = \frac{289}{36}$ ,  $\sqrt{D} = \frac{17}{6}$ .

$$\lambda_1 = -\frac{3}{2}, \quad \lambda_2 = \frac{4}{3}.$$

Both  $|\lambda_1|, |\lambda_2| > 1$ : **stationary**.

## Example 2: AR(2), Non-Stationary

### Example

$$X_n = \frac{1}{3}X_{n-1} + \frac{2}{3}X_{n-2} + Z_n.$$

Characteristic equation:  $2\lambda^2 + \lambda - 3 = 0$ ,  $D = 25$ .

$$\lambda_1 = -\frac{3}{2}, \quad \lambda_2 = 1.$$

$|\lambda_1| > 1$  but  $|\lambda_2| = 1 \not\neq 1$ : **not stationary**.

### Example 3: AR(2), Complex Roots

#### Example

$$X_n = 2X_{n-1} - 2X_{n-2} + Z_n.$$

Characteristic equation:  $2\lambda^2 - 2\lambda + 1 = 0$ ,  $D = 4 - 8 = -4 < 0$  (complex roots):

$$\lambda_{1,2} = \frac{2 \pm \sqrt{-4}}{4} = \frac{1}{2} \pm \frac{1}{2}i.$$

$$|\lambda_{1,2}| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \frac{\sqrt{2}}{2} < 1.$$

**Not stationary.**

## Autoregressive Model AR( $p$ )

$$X_n = \mu + \alpha_1(X_{n-1} - \mu) + \cdots + \alpha_p(X_{n-p} - \mu) + Z_n.$$

### Example

E.g. AR(3):  $X_n = aX_{n-1} + bX_{n-2} + cX_{n-3} + Z_n$ . Given  $a, b, c, X_0, X_1, X_2$ , generate  $Z_3$  to get  $X_3 = aX_2 + bX_1 + cX_0 + Z_3$ , then  $X_4 = aX_3 + bX_2 + cX_1 + Z_4$ , and so on.

### Definition

The characteristic equation  $1 - \alpha_1\lambda - \cdots - \alpha_p\lambda^p = 0$  has  $p$  roots  $\lambda_1, \dots, \lambda_p$  (possibly complex). The AR( $p$ ) series is stationary iff  $|\lambda_1| > 1, \dots, |\lambda_p| > 1$ .

The roots may be found numerically (e.g. in Python) for larger  $p$ .

## Moving Averages MA( $q$ )

### Definition

$$X_n = \mu + Z_n + \beta_1 Z_{n-1} + \cdots + \beta_q Z_{n-q}.$$

Since  $\mathbb{E}(Z_{n-j}) = 0$  for all  $j$ ,  $\mathbb{E}(X_n) = \mu$  — constant. To check the covariance condition, take  $q = 1$ ,  $\mu = 0$ :  $X_n = Z_n + \beta Z_{n-1}$ ,  $X_{n+m} = Z_{n+m} + \beta Z_{n+m-1}$  ( $m > 0$ ). Expanding,

$$\text{cov}(X_n, X_{n+m}) = \underbrace{\text{cov}(Z_n, Z_{n+m})}_{=0} + \beta \underbrace{\text{cov}(Z_{n-1}, Z_{n+m})}_{=0} + \beta \underbrace{\text{cov}(Z_n, Z_{n+m-1})}_{=\sigma^2 \text{ if } m=1, 0 \text{ if } m>1} + \beta^2 \underbrace{\text{cov}(Z_{n-1}, Z_{n+m-1})}_{=0}$$

— depends on  $m$  only.

### Remark

In other words, MA( $q$ ) is always stationary.

## Full ARMA( $p, q$ ) Model

### Definition

“AR” = autoregressive, “MA” = moving average — this model includes past white noise too:

$$X_n = \mu + \alpha_1(X_{n-1} - \mu) + \cdots + \alpha_p(X_{n-p} - \mu) + Z_n + \beta_1 Z_{n-1} + \cdots + \beta_q Z_{n-q}.$$

### Remark

The characteristic equation and stationarity condition for ARMA( $p, q$ ) coincide with those of its AR( $p$ ) component — **you may ignore the  $\beta$ 's** for stationarity.

## The ARIMA( $p, d, q$ ) Model: Motivation

Consider an ARMA(2, $q$ ) model with characteristic equation  $1 = \alpha_1 \lambda + \alpha_2 \lambda^2$ . Suppose  $\lambda = 1$  is a root: then  $|\lambda| \not> 1$ , so  $\{X_n\}$  is **not stationary**.

Substituting  $\lambda = 1$ :  $1 = \alpha_1 + \alpha_2$ , so  $\alpha_1 = 1 - \alpha_2$ . Then

$$X_n = \alpha_1 X_{n-1} + \alpha_2 X_{n-2} + Z_n + \cdots = (X_{n-1} - \alpha_2 X_{n-1}) + \alpha_2 X_{n-2} + Z_n + \cdots$$

$$X_n - X_{n-1} = -\alpha_2 (X_{n-1} - X_{n-2}) + Z_n + \cdots.$$

Denoting  $Y_n := X_n - X_{n-1}$  (the **differencing** of  $X_n$ ):  $Y_n = -\alpha_2 Y_{n-1} + Z_n + \cdots$  — an ARMA(1, $q$ ) series. If  $|\alpha_2| < 1$ ,  $Y_n$  is stationary.



## ARIMA( $p, d, q$ ): Definition

### Definition

If the characteristic equation of an ARMA(2, $q$ ) model has roots  $\lambda_1 = 1$  and  $|\lambda_2| > 1$ , then  $\{X_n\}$  is not stationary, but  $Y_n = X_n - X_{n-1}$  is. We then say  $\{X_n\}$  follows an **ARIMA(1,1, $q$ )** model — the first “1” because  $Y_n$  is ARMA(1,0)-like, the second “1” because differencing was applied once.

**Recovering  $X_n$  from  $Y_n$ :**

$$Y_n = X_n - X_{n-1} \iff X_n = Y_n + X_{n-1} = Y_n + Y_{n-1} + X_{n-2} = \cdots = Y_n + Y_{n-1} + \cdots + Y_1 + X_0.$$